



SKILL ENHANCEMENT COURSE

Assignment - 1

Name : Sshyamolika Gungun

Sejal Shreya

Saadgi Puniwala

Class : 4C

USN : ENG24DS0190

ENG24DS153

ENG24DS148

Power BI Project Report

Netflix Shows and Movies Analysis

1. Problem Statement & Objective

The objective of this project is to analyze Netflix's dataset of movies and TV shows to identify patterns in content distribution, growth trends, and category preferences.

The goal is to:

- Understand how Netflix content has grown over time
- Analyze the distribution of Movies vs TV Shows
- Identify popular genres/categories
- Provide insights for content strategy and decision-making

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	show_id	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	type						
2	81193313	Chocolate		Ha Ji-won, South Korea	November	2019	TV-14	1 Season		International	Brought to TV Show							
3	81197050	Guatemala	Luis Ara, Ig	Christian Morales	November	2019	TV-G	67 min		Document	From Sierra	Movie						
4	81213894	The Zoya F	Abhishek S	Sonam Kap	India	November	2019	TV-14	135 min		Comedies, A	goofy cc	Movie					
5	81082007	Atlantics	Mati Diop	Mama San	France, Senegal	November	2019	TV-14	106 min		Dramas, In	Arranged t	Movie					
6	80213643	Chip and Potato		Abigail Ol	Canada, United Kingdom	2019	TV-Y	2 Seasons		Kids' TV	Lovable p	TV Show						
7	81172754	Crazy peo	Moses Inw	Ramsey N	Nigeria	November	2018	TV-14	107 min		Comedies,	Nollywood	Movie					
8	81120982	I Lost My f	JÃ©rÃ©m	Hakim Fari	France	November	2019	TV-MA	81 min		Dramas, In	Romance, Movie						
9	81227195	Kalushi: Th	Mandla Du	Thabo Rari	South Africa	November	2016	TV-MA	107 min		Dramas, In	The life an	Movie					
10	70205672	La Reina del Sur		Kate del C	United States, Spain, Mexico	2019	TV-14	2 Seasons		Crime TV S	This comp	TV Show						
11	81172841	Lagos Real	Mike Ezur	Nonso Diobi, Mike Ez	Nigeria	November	2018	TV-14	118 min		Comedies,	Two mooc	Movie					
12	81172899	Payday	Cheta Chu	Baaj Adeb	Nigeria	November	2018	TV-MA	110 min		Comedies,	After an e	Movie					
13	81094391	Sugar Rush	Christmas	Hunter Ma	United States	November	2019	TV-PG	1 Season		Reality TV	It's everyt	TV Show					
14	81172908	The Accide	Roger Rus	Ramsey N	Nigeria	November	2017	TV-14	104 min		Action & A	Nursing a	TV Show					
15	81152641	The Charming	Stepm	Shahkrit Yamnarm, Vi	Nigeria	November	2019	TV-14	1 Season		International	A quirky fa	TV Show					
16	81172901	The Island	Toka McB	Sambasa N	Nigeria	November	2018	TV-14	93 min		Dramas, In	When a cc	Movie					
17	80990849	The Movies	That Made	Us	United States	November	2019	TV-MA	1 Season		Docuseries	These bloc	TV Show					
18	81033086	Holiday Ru	Leslie Sma	Romany M	United States	November	2019	TV-PG	94 min		Children & A	widower	Movie					
19	80156799	Leviath		Nobunaga	Japan	November	2019	TV-14	1 Season		Anime Seri	Young Levi	TV Show					
20	81161538	Lugar de Mulher			Brazil	November	2019	TV-MA	1 Season		International	Four come	TV Show					
21	80997965	Merry Happy	Whatever	Dennis Qui	United States	November	2019	TV-PG	1 Season		TV Comed	A strugglin	TV Show					
22	81018979	Mythomaniac		Marina Ha	France	November	2019	TV-MA	1 Season		International	Burned ou	TV Show					
23	60020826	The Score	Frank Oz	Robert De	Germany, United States	November	2001	R	124 min		Dramas, T	Ready-to-i	Movie					
24	81002391	Broken			United States	November	2019	TV-14	1 Season		Docuseries	This invest	TV Show					
25	70101696	Defiance	Edward Zv	Daniel Cra	United States	November	2008	R	137 min		Action & A	In this acti	Movie					
26	81194544	Evvarikee	Basava Shi	Rakesh Va	India	November	2019	TV-14	134 min		Comedies,	When cast	Movie					
27	80995081	Little Sine	Prakash Sa	Saumba Daan, Sonal N	Nigeria	November	2019	TV-Y7	69 min		Children & In	a iourne	Movie					

2. Dataset Description

- **Dataset Name:** Netflix Titles Dataset
- **Source:** Kaggle (Netflix Shows and Movies Dataset)
- **Size:** 5000+ records

Key Features (Columns):

- `show_id` – Unique ID
 - `title` – Name of show/movie
 - `director` – Director name
 - `cast` – Actors involved
 - `country` – Country of origin
 - `date_added` – Date added to Netflix
 - `release_year` – Year of release
 - `rating` – Content rating (TV-MA, PG, etc.)
 - `duration` – Duration of content
 - `listed_in` – Genre/category
 - `type` – Movie or TV Show
-

3. Data Cleaning & Preprocessing

The following steps were performed in Power BI:

- Removed missing/null values where necessary
- Converted `date_added` to proper date format
- Extracted **Year** from `date_added` for trend analysis
- Ensured correct data types (text, number, date)
- Standardized category values in `listed_in`

Dashboard & Visualizations Explanation

You created the following visuals:

1. Line Chart – Growth Over Years

- Shows count of titles by release year
- Insight: Rapid growth after 2010, especially post-2015

2. Bar Chart – Top Categories

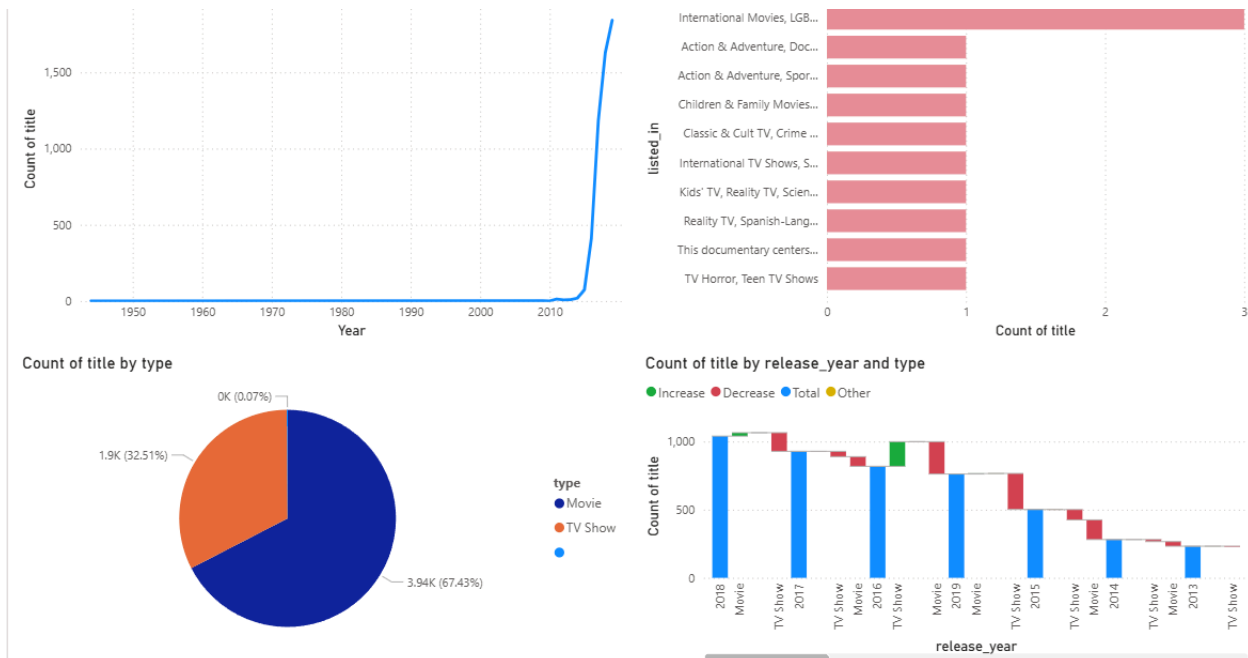
- Displays top genres using `listed_in`
- Insight: Categories like Drama, Comedy, and Action dominate

3. Pie Chart – Content Type Distribution

- Movies vs TV Shows
- Insight: Movies (~67%) dominate over TV Shows (~33%)

4. Waterfall Chart – Year-wise Contribution

- Shows increase/decrease in content count per year
- Insight: Significant rise in recent years, slight fluctuations



5. Key Insights & Conclusions

- Netflix content has **grown significantly after 2015**
 - Movies dominate the platform compared to TV Shows
 - Popular genres include:
 - Drama
 - Comedy
 - Action & Adventure
 - There is a **consistent increase in content production** over time
 - Recent years show aggressive content expansion
-

6. Tools & Techniques Used

- **Power BI Desktop**
 - Data Cleaning using Power Query
 - Data Visualization techniques
 - Basic Data Modeling
 - DAX for calculations
-

7. Data Model Design

- Single table dataset used
 - No complex relationships required
 - Structure follows a **flat table model**
 - Suitable for exploratory analysis
-

8. DAX Measures & Calculated Columns

Measures Used:

- `Count of Title = COUNT(title)`

Calculated Columns:

- Extracted Year:
Year = YEAR(date_added)
-

- **Month column**

Month = FORMAT(date_added, "MMMM")

Content Age

Content Age = YEAR(TODAY()) - release_year

9. KPI Definitions & Business Metrics

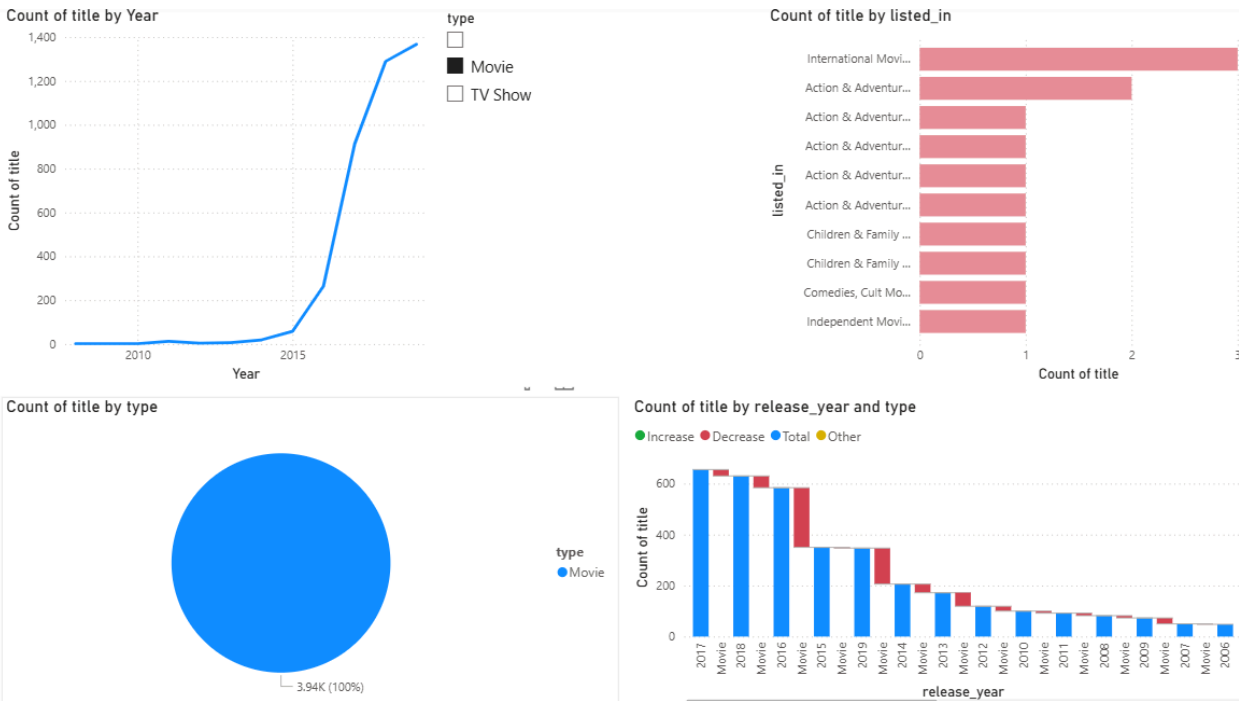
- Total Titles (Movies + TV Shows)
 - Total Movies
 - Total TV Shows
 - Content Growth by Year
 - Top Genres
-

10. Filters / Slicers & Interactivity

Basic filters were applied at both the visual and page levels to refine the analysis and focus on relevant data points. In addition, a slicer was implemented to enhance interactivity within the dashboard.

The slicer, based on the **content type (Movie/TV Show)**, allows users to dynamically filter and explore the dataset. By selecting different options in the slicer, all visuals in the dashboard update automatically, enabling a more flexible and user-driven analysis.

This interactivity improves the usability of the dashboard and helps in better understanding patterns and trends within specific categories of content. Although advanced interactive features such as map visuals were not included, the use of slicers significantly enhances the overall analytical experience.



11. Performance Considerations

- Dataset size is moderate (5000+ rows)
- No performance issues observed
- Optimization steps:
 - Removed unnecessary columns
 - Used simple measures

12. Challenges Faced

- Handling missing values in some columns
- Understanding proper filter types (Top N, basic, advanced)
- Choosing appropriate visuals (bar vs waterfall vs ribbon)

Solutions:

- Cleaned data in Power Query
- Used correct filter settings
- Selected visuals based on data meaning

13. Business Recommendations

- Focus more on **popular genres** like Drama and Comedy
 - Increase production of **TV Shows** (less than Movies)
 - Continue investment in content as growth trend is strong
 - Expand globally based on country-wise data (future scope)
-

14. Limitations

- Dataset is limited to available Kaggle data
 - No real-time updates
 - No user ratings or revenue data included
-

15. Future Scope / Enhancements

- Add **Slicers** for better interactivity
 - Include **Map Visual** for country analysis
 - Perform **sentiment analysis on descriptions**
 - Add advanced KPIs and dashboards
 - Use **Ribbon Chart for ranking trends**
-

Conclusion

This project successfully analyzes Netflix's content dataset to uncover trends in growth, distribution, and genre popularity. The findings provide valuable insights into the platform's content strategy and demonstrate effective use of data visualization techniques in Power BI.